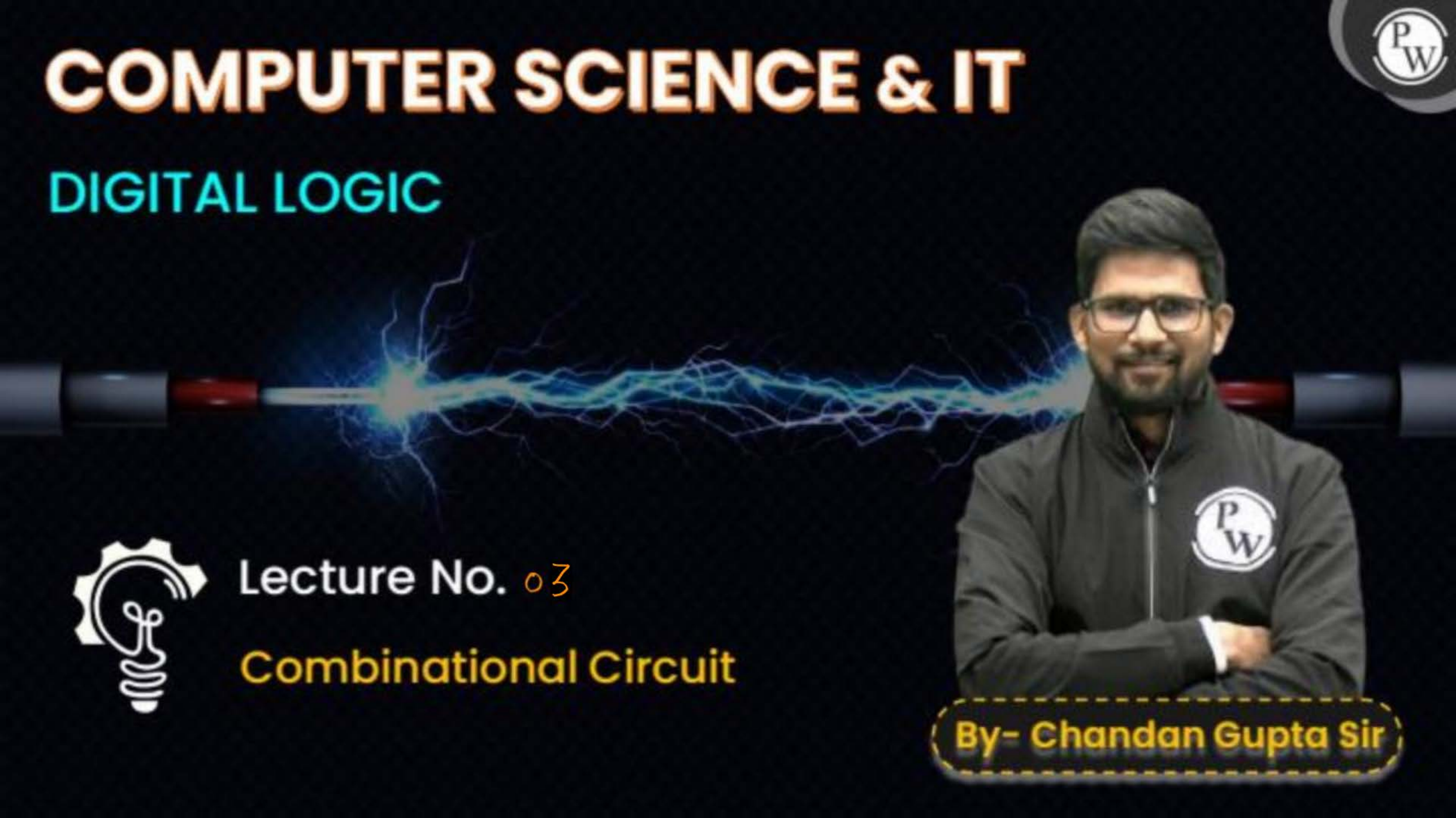
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 03

Combinational Circuit

By- Chandan Gupta Sir



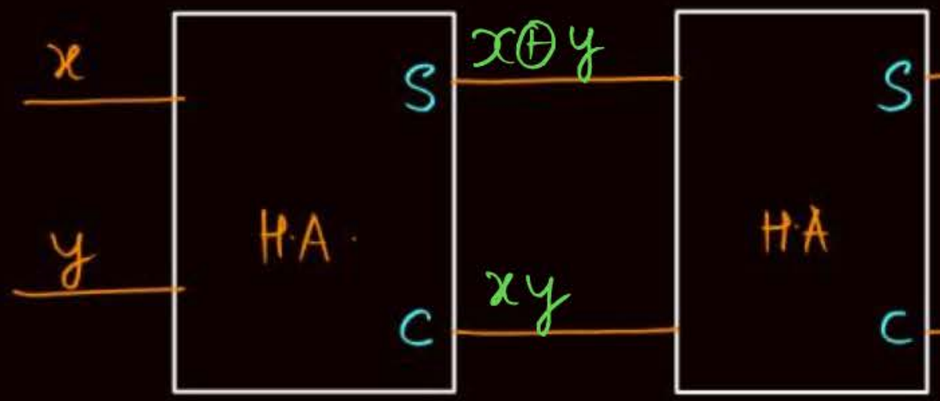
Comb. CKt.



- Serial Adder
- Parallel adder

H.W

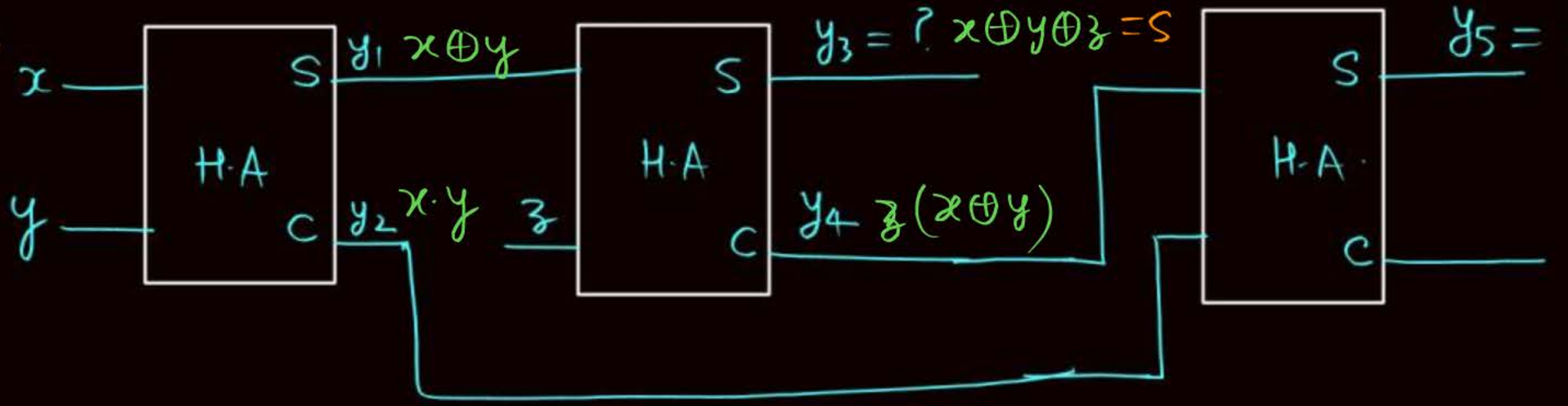
Q.1



$y_1 = ? = x \oplus y \oplus xy = x + y$

$y_2 = ? = (x \oplus y) \cdot xy = (\bar{x}y + x\bar{y})xy = 0$

Q.2



$$y_5 = [z.(x \oplus y)] \oplus xy = [zx \oplus zy] \oplus xy = xy \oplus yz \oplus xz$$

$$= [z(\bar{x}y + x\bar{y})] \oplus xy$$

$$= \underbrace{(\bar{x}yz + x\bar{y}z)}_{\Sigma 3,5} \oplus \underbrace{xy}_{\Sigma(6,7)}$$

$$= \Sigma(3,5,6,7)$$

xy	
1 1	0
1 1	1

$$= \Sigma(3,5,6,7) \qquad = xy + yz + zx$$

$$= xy + (x \oplus y)z$$

$$xy \oplus yz \oplus xz = xy + yz + zx$$

Adders

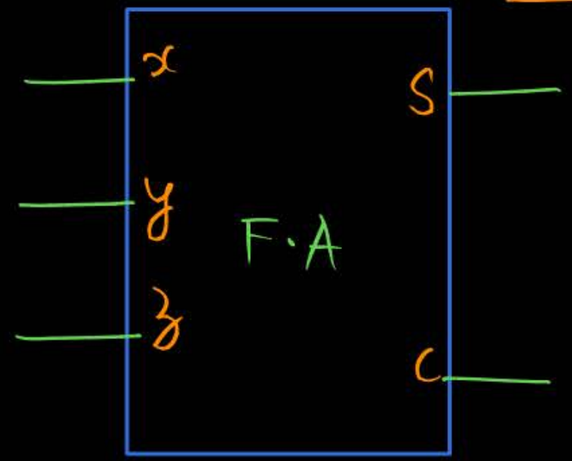
- Serial Adder
- Parallel Adder
- Look ahead carry adder

$A = a_3\ a_2\ a_1\ a_0$

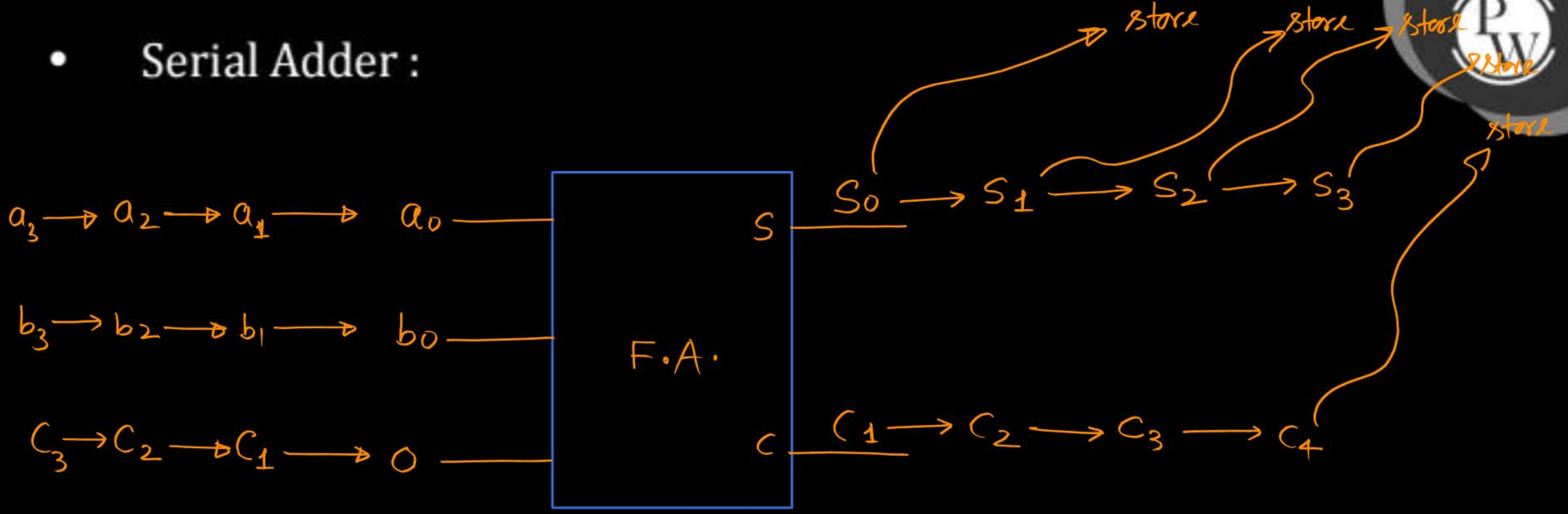
$B = b_3\ b_2\ b_1\ b_0$

$+\ C_3\ C_2\ C_1\ 0$

Result → $C_4\ S_3\ S_2\ S_1\ S_0$

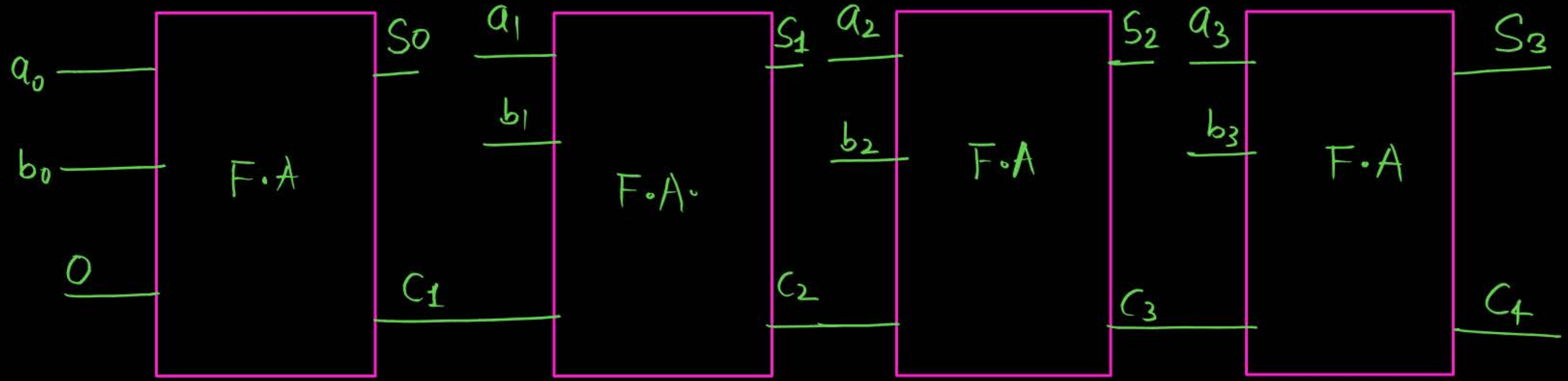


- **Serial Adder :**



To add n-bit numbers in serial adder, we require only one F.A. Time taken (delay) to generate the complete result is high due to serial operation in serial adder.

- Parallel Adder :



Ripple carry adder.

Case-I

Sum delay of each F.A $t_s = 2\,ns$

Carry delay of each F.A $t_c = 1.5\,ns$

	4.5ns	3ns	1.5ns	
	a_3	a_2	a_1	a_0
	b_3	b_2	b_1	b_0
+	C_3	C_2	C_1	0
	C_4	S_3	S_2	S_1	S_0
	6ns	6.5ns	5ns	3.5ns	2ns

Worst case (over all delay) $T_d = 6.5\,ns$

Case-II

$t_s = 1\,ns$

$t_c = 1.5\,ns$

	4.5ns	3ns	1.5ns	
	a_3	a_2	a_1	a_0
	b_3	b_2	b_1	b_0
	C_3	C_2	C_1	0
	C_4	S_3	S_2	S_1	S_0
	6ns	5.5ns	4ns	2.5ns	1ns

Worst case (overall delay) $T_d = 6\,ns$

Worst case delay $T_d = (n-1)t_c + \max[t_c, t_s]$

Worst case delay $T_d = [t_{c1} + t_{c2} + \cdots t_{c(n-1)}] + \max[t_{cn}, t_{sn}]$

- $T_d \rightarrow$ represents the time taken to perform one complete addition of two n-bits numbers.

no. of additions in 1 sec $= \dfrac{1 \text{ sec}}{T_d(\text{sec})}$



2 Minute Summary

- Serial adder
- Parallel adder.

Thank you

GW

Soldiers !

